

What is a plant?

PLANTS ARE THE KEY to life on Earth. Without them, a whole host of other organisms would soon disappear. This is because many living things depend on plants for food and for somewhere to live. Plants are different, because they make food themselves, using energy that they collect from sunlight. There are two main kinds of plant. The simplest ones do not have flowers. They include mosses, liverworts, ferns, and also conifers—the biggest plants that have ever existed on Earth. Compared to them, flowering plants are newcomers, but they have become a huge success. Today, there are about a quarter of a million different kinds of flowering plant, and they grow almost everywhere, from mountaintops to deserts. This book tells their story.



THIS IS NOT A PLANT
It is often difficult to tell simple plants and animals apart. This plantlike organism is a hydrozoan and lives in the sea. Its branches are formed by tiny animals called polyps, which have tentacles to trap particles of food.



THIS IS A PLANT
A lichen is made up of two different organisms: a tiny nonflowering plant called an alga, and a fungus. The algal cells live among the tiny threads formed by the fungus and supply the fungus with food, which they make using sunlight (pp. 14–15). The fungus cannot make its own food and would die without the alga. Lichens grow very slowly and are extremely long-lived.

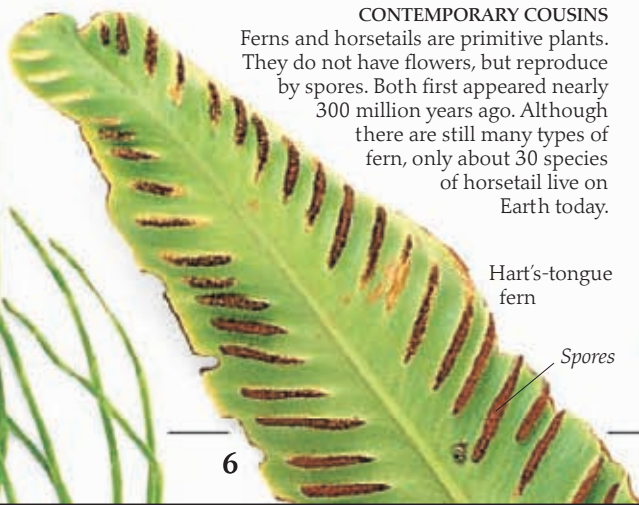
Lichens growing on limestone rock



THIS WAS A PLANT
Forests of horsetails and giant club-mosses, up to 150 ft (45 m) tall, once formed a large part of the Earth's vegetation (see right). Over 300 million years, their buried remains have gradually turned into coal.

Horsetail

CONTEMPORARY COUSINS
Ferns and horsetails are primitive plants. They do not have flowers, but reproduce by spores. Both first appeared nearly 300 million years ago. Although there are still many types of fern, only about 30 species of horsetail live on Earth today.



Hart's-tongue fern

Spores